

Rahul Kumar

About

I am a Computer Science engineering student from [BIT_Sindri](#). Expertise in advance data structures, Artificial Intelligence, algorithms and Machine Learning. And Interested in theoretical details and implementation. With a good knowledge in Statistics and Data Science. Also familiar with basic web programming and GUI.

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Education

Institution	Education	Year
MITx	MicroMasters in Statistics and Data Science	2021-22
BIT Sindri	B.Tech Computer Science	2018-22
Guru Gobind Singh Public School	Higher Secondary Education	2015-17
ARS Public School	High School	2008-15
Sharswati Sishu Vidya Mandir	Elementary School	2005-08

Skills

- **Advance Data Structures**
- **Algorithms**
- **Artificial Intelligence**
- Search
- CSP
- Optimization
- RL
- Knowledge Based Agents
- **Statistics**
- Estimation
- Hypothesis Testing
- **Probability Theory**
- Probabilistic Modeling
- Inference
- **Machine Learning**
- Clustering
- Recommendation System
- Reinforcement Learning
- Deep Learning
- TensorFlow
- PyTorch
- Scikit-Learn
- Data visualization
- **Languages**
- Python
- C
- C++
- JavaScript
- **Web Programming**
- HTML
- CSS
- Flask
- Django
- Saas
- Bootstrap
- **DBMS**
- SQL
- **Arduino**
- **Competitive Programming**
- **Tools**
- Git
- heroku
- google cloud
- jinja

Achievements

1. **Implemented van Emde Boas Tree**
1st time correct implementation in python in the History [Visit](#)
2. **World Programming Championship - IIT Mumbai**
Rank 69 | Certificate no 99695
3. **Gold in HackFest 2020**
Rank 29 | 8051 Rating 2042.08 from 1500 [Visit](#)
4. **GATE 2021**
Got AIR 5648 [Visit](#)
5. **CodeVita Season 9**
Cleared Round 1 with Rank 1042
6. **CodeVita Season 9**
Got International 242 in Round 2
7. **Hack the Interview III**
Rank(73 | 919) [Visit](#)
8. **Hack the Interview IV (Asia Pacific)**
Rank(100 | 4353) [Visit](#)
9. **Hack the Interview IV (U.S.)**
Rank(65 | 1217) [Visit](#)
10. **Hackerrank Algorithms**
Rank 815/ 2219538 [Visit](#)
11. **Hackerrank Data Structures**
Rank 169| 931788 [Visit](#)
12. **Hackerrank Python Practice**
Rank 1 [Visit](#)

Certifications

- [MIT Fundamentals of Statistics](#)
- [MIT Probability - The Science of Uncertainty](#)
- [MIT Machine Learning with Python-From Linear Models to Deep Learning](#)
- [CS50 Introduction to Artificial Intelligence with Python](#)
- [IBM Introduction to AI](#)
- [IBM Python for Data Science, AI & Development](#)
- [Udemy Complete 2020 Data Science & Machine Learning Bootcamp](#)
- [C Advanced](#)
- [C Basic](#)
- [C Intermediate](#)
- [CSS](#)
- [Java](#)
- [Kaggle Intermediate Machine Learning](#)
- [Kaggle Intro to Deep Learning](#)
- [Kaggle Intro to Machine Learning](#)
- [Kaggle Python](#)
- [Problem Solving Advanced](#)
- [Problem Solving Basic](#)
- [Problem Solving Intermediate](#)
- [Python Advanced](#)
- [Python Basic](#)
- [Python Intermediate](#)
- [SQL Basic](#)
- [SQL Intermediate](#)
- [Udemy Data Structures and Algorithms](#)
- [Udemy Python](#)

Projects

1. **PacMan AI**
AI based intelligent agent to control pacman in different environments. The pacman plays against the ghosts and tries to eat all food while staying safe from ghosts and finishing in minimum time. The pacman uses different methods like A* Search, Heuristics, Reinforcement Learning with different parameters.
2. **Automatic Review Analyzer**
Sentiment Analysis for customer reviews. Using simple perceptron algorithm
3. **MNIST Digit Recognition**
Classification of digits using support vector machine and gradient descent.
4. **Overlapping Digit Recognition**
Classification of overlapping digits which contain multiple digits in single image, where a digit is written over other digit, using Convolutional Neural Networks.
5. **Collaborative Filtering via Gaussian Mixtures**
Recommender system for Netflix using Gaussian Mixtures models and EM algorithm. Ratings of few users for different movies are provided as input and ratings for movies for which user has not rated is calculated. It achieves very nice accuracy.
6. **Text game using Reinforcement Learning**
It is a game in which a robot is given task in a room and he learns to play it. Implementations using Q learning, Q learning with approximations, using neural networks to reduce the exponential size Q Tables.
7. **Maze Solver**
Demonstration of complexity, speed, effectiveness of different search algorithms
8. **Degrees**
Calculation of minimum degree of separation between actors in holywood. Separation is in terms of working together.
9. **Tic-Tac-Toe AI**
MiniMax and Alpha-Beta pruning based undefeatable Intelligent to play against human.
10. **Minesweeper AI**
Knowledge based Intelligent agent to play Minesweeper which reasons using prepositional logic.
11. **CSP solver for Crossword Puzzle**
Solves Crossword puzzles by modelling it into Constraint satisfaction problem and then using backtracking search with different heuristics to improve performance
12. **Pabilistic models for reasoning under uncertainty**
To demonstrate page ranking and gene inheritance, by sampeling and using bayes-net
13. **Shopping**
K Nearest Neighbour model to predict whether customer will make the purchase or not.
14. **NIM**
Playing NIM against human. Trained using reinforcement learning.
15. **Traffic Sign classification**
CNN model for 43 class classification of traffic sign boards
16. **CS50 Wiki**
Mini version of WikiPedia with almost all features of wikipedia. Built using django, html, jinja, css, saas.
17. **HomePage**
My Homepage which contains details about me like hobbies, interests, and many things more. Built using flask, HTML, CSS, Javascript.
18. **CS50-Finance**
A lite app to get quotes of stocks and to perform buy, sell stocks. Technologies used: Flask, SQL, APIs, JSON, Heroku, GIT.

Hobbies

- Digital Arts
- Tabla Playing